

State of Health Estimation Using a Novel Dataset Under Extreme Temperature and Aging Conditions

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Abstract—WIP

Index Terms—WIP

I. INTRODUCTION

WIP

II. DATA PROCESSING

A. Dataset

The dataset used was obtained from the Scienlab SL1007A charge-discharge testing platform, where eight Samsung INR21700-50E lithium-ion cells were tested from fresh SoH levels to beyond EOL. Telemetry of each cell was recorded using Measurement and Control Units (MCU1–MCU8) and exported using the ASAM Measurement Data Format (MF4). Cell specifications are given in Table I.

TABLE I
CELL SPECIFICATIONS

Parameter	Value
Nominal capacity (C_{nom})	4.9 Ah
Upper cut-off voltage (U_{max})	4.2 V
Lower cut-off voltage (U_{min})	2.5 V
Max. discharge current	−20 A (−4.0C)
Max. charge current	8 A (2.0C)
Temperature range	−20 to 60 °C
State of charge range	0 to 100 %

B. Test Protocols

Each cell underwent four discharge protocols across various ambient temperatures: Constant, Pulse, Hybrid Pulse Power Characterization (HPPC), and the Worldwide Harmonised Light Vehicles Test Cycle (WLTC) (Fig. 1). Testing was divided into three sequential phases (Table II):

- 1) **Initial measurements:** Baseline testing on fresh cells across all four protocols.
- 2) **Accelerated aging:** Cyclic aging using constant 1C discharge cycles at 25°C.
- 3) **Post-aging measurement:** Re-testing using the baseline protocols.

Temperature band distribution across phases is given in Table III.

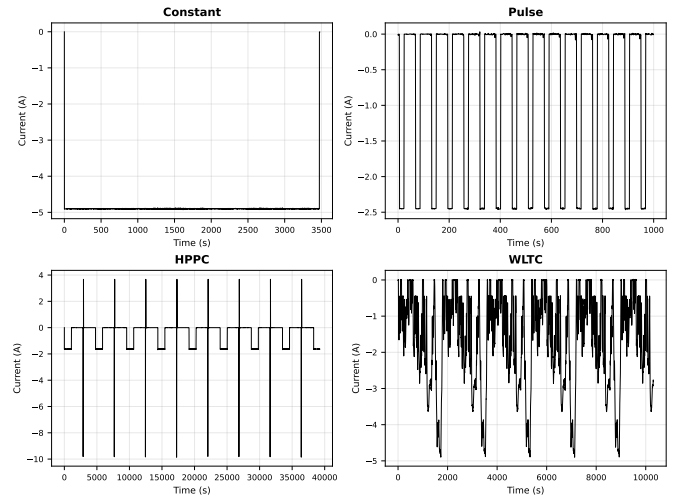


Fig. 1. Discharge profiles for the four test protocols: Constant, Pulse, HPPC, and WLTC.

TABLE II
SOH, C-RATES, AND PROTOCOL DISTRIBUTION

Phase	SoH Range (%)	Current Rates	Protocols
Initial	98 – 100	0.5C to 4C	Constant
			Pulse
			HPPC
			WLTC
Aging	70 – 98	1C	Constant
Post-Aging	< 70 – 80	0.5C to 4C	Constant
			Pulse HPPC

TABLE III
TEMPERATURE BAND DISTRIBUTION

Phase	Temperature Bands (°C)						
	−20	−10	−5	0	10	25	40
Initial	38	30	6	74	14	127	59
Aging	0	0	0	0	0	792	0
Post-Aging	39	50	0	54	8	95	56
Total	77	80	6	128	22	1014	115

C. Pre-processing Pipeline

As per the dataset engineer’s advice, voltage (V), current (A), temperature (T), and ambient temperature (T_{amb}) are primary channels recorded from the test platform’s sensors. The components of the pre-processing pipeline are as follows:

- 1) **Extract discharge periods:** Using discharge start/end signals provided by the testing platform to filter only discharge periods.
- 2) **Hierarchical Data Format (HDF5):** Converting each discharge period away from MF4 to HDF5 allows for O(1) disk reads when loading the dataset, while also only taking primary channels (V, I, T, T_{amb}) to reduce memory usage.
- 3) **Static conditions metadata:** T_{amb} is controlled by the test platform at a fixed value for each discharge period, thus an average value was taken and stored as an HDF5 attribute.
- 4) **Statistics metadata:** Statistics (mean, M_2 , bounds) are stored as HDF5 attributes for standardization.
- 5) **SoH metadata:** SoH is calculated for each discharge period and written to HDF5 attributes (detailed below in Section D).

D. SoH Labeling

Raw SoH for each discharge period was computed via Coulomb counting:

$$\text{SoH}_{\text{raw}} = \frac{\int_{t_{\text{start}}}^{t_{\text{end}}} |I(t)| dt}{C_{\text{nom}}} \quad (1)$$

Because measured capacity varies with discharge rate and T_{amb} , reference points were filtered from 1C discharge cycles at 25°C. A 4th degree polynomial was then fitted over the cycle indices to establish SoH trajectories for each MCU (Fig. 2). Reference-only SoH ranges and cycle counts are found in Table IV.

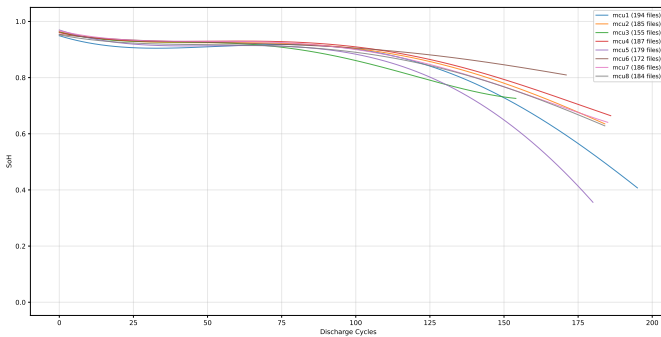


Fig. 2. Fitted SoH trajectories for all eight MCUs across their full discharge cycle histories.

TABLE IV
MCU SOH RANGE AND CYCLE COUNT

MCU	SoH Range (%)	Cycles
1	94.3–47.9	194
2	95.6–67.8	185
7	95.8–67.5	186
4	95.3–71.1	187
5	95.7–42.3	179
6	95.8–82.7	172
3	97.4–74.0	155
8	94.6–66.8	184

E. Statistics

Global dataset statistics are computed across all $N = \sum N_i$ time steps using the pre-aggregated file metadata ($N_i, \mu_i, M_{2,i}$) for normalization:

$$\mu = \frac{\sum_i N_i \mu_i}{N}, \quad \sigma = \sqrt{\frac{\sum_i M_{2,i} + \sum_i N_i \mu_i^2 - N \mu^2}{N}} \quad (2)$$

For scalar attributes, sample-wise mean and variance are calculated across all K discharge instances (Table V).

TABLE V
FEATURE STATISTICS

Feature	Mean (μ)	Std. Dev. (σ)
Voltage (V)	3.5997	0.3253
Current (I)	-1.5349	2.1722
Cell Temperature (T)	16.9000	18.0783
Ambient Temperature (T_{amb})	15.4799	18.0773
Thermal Delta (ΔT_{cell})	5.0178	3.3843
State of Health (SoH)	0.8578	0.1029

III. METHODOLOGY

A. Problem Formulation

Given a sequence of V , I , and T over a discharge period, conditioned on T_{amb} , an estimator function $f(\cdot)$ predicts the cell’s SoH:

$$\widehat{\text{SoH}} = f(V, I, T \mid T_{\text{amb}}) \quad (3)$$

B. Estimator Architecture

Input sequences pass through three 1D Convolutional Neural Network (1D-CNN) blocks (32, 64, 128 channels; GroupNorm, GELU, dropout), followed by Feature-wise Linear Modulation (FiLM) conditioning on T_{amb} :

$$\text{FiLM}(h) = \gamma(T_{\text{amb}}) \odot h + \beta(T_{\text{amb}}) \quad (4)$$

where γ and β are learned affine parameters. The modulated features pass through a 2-layer bidirectional Gated Recurrent Unit (BiGRU), whose final hidden states are concatenated and projected to a scalar SoH estimate.

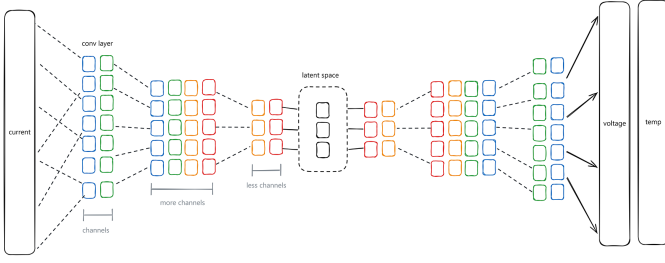


Fig. 3. Model architecture: 1D-CNN blocks, BiGRU, and FiLM conditioning on T_{amb} .

C. Training Setup

For baseline testing, the dataset was sliced into 2000 timestep windows (approx. 30 minutes) with a stride of 500. The MCUs were organized into a 5, 1, 1 split for the training, validation and testing sets respectively. Additionally, normalization was applied to all input sequences and conditions. Estimator hyperparameters are given in Table VI.

TABLE VI
ESTIMATOR HYPERPARAMETERS

Hyperparameter	Value
Conv. base channels	32
Conv. kernel size	7
Conv. stride	2
GRU hidden size	128
GRU layers	2
Dropout	0.1
Batch size	512
Epochs	500
Learning rate	0.0005

D. Baseline Results and Discussion

Evaluation was performed on the validation and testing sets across SoH ranges (Table VII) and across temperature bands and discharge protocols (Table VIII).

TABLE VII
BASELINE ESTIMATOR PERFORMANCE

Phase	SoH	RMSE	MAE	R^2	%Err	Cycles
Initial	91.8–96.2	0.016	0.012	−0.71	1.2%	65
Aging	78.1–93.4	0.013	0.011	0.89	1.2%	197
Post-Aging	62.8–82.0	0.034	0.028	0.50	3.8%	77
Overall	62.8–96.2	0.020	0.015	0.94	1.8%	339

R^2 is negative for Initial (−0.94) and Post-Aging (−1.16), an artifact of their narrow SoH windows (92.5–97.4% and 74.0–82.5%). With little target variance, R^2 amplifies residual error

and is uninformative, so RMSE and %Err are the meaningful metrics. Initial achieves 1.4% error (RMSE 0.018). Aging is the strongest phase ($R^2 = 0.86$, 1.3% error, RMSE 0.015): its single 1C constant discharge shape gives a consistent waveform the CNN/BiGRU exploits while still tracking declining SoH across discharges. Post-Aging error rises to 3.9% (RMSE 0.039) because the MCUs’ SoH trajectories diverge after accelerated aging, producing a sparser distribution less represented in training. Overall, $R^2 = 0.88$ and 1.7% error span 74.0–97.4% SoH.

TABLE VIII
ESTIMATOR PERFORMANCE BY TEMPERATURE AND PROTOCOL

Evaluation Slice	SoH (%)	RMSE	MAE	R^2	%Err	Cycles
Temperature Bands						
−20°C	68.1–93.7	0.024	0.020	0.94	2.7%	16
−10°C	62.8–93.1	0.037	0.030	0.86	4.3%	18
−5°C	92.4–92.4	0.003	0.003	—	0.3%	1
0°C	64.2–94.1	0.029	0.021	0.93	2.8%	28
10°C	69.3–92.5	0.007	0.007	0.99	0.9%	4
25°C	65.5–96.2	0.017	0.013	0.91	1.5%	247
40°C	66.8–95.1	0.019	0.015	0.96	1.9%	25
Discharge Protocols						
Constant	62.8–96.2	0.021	0.015	0.92	1.8%	246
HPPC	68.1–92.6	0.011	0.009	0.99	1.2%	23
Pulse	71.0–95.1	0.020	0.015	0.95	1.9%	67
WLTC	92.1–95.3	0.014	0.012	−0.11	1.3%	3
Overall Dataset	62.8–96.2	0.020	0.015	0.94	1.8%	339

The estimator performs consistently at both ends of the SoH spectrum within each temperature band. For example, −20°C spans 74.0–96.4% at $R^2 = 0.93$ and 40°C spans 75.6–97.4% at $R^2 = 0.95$, indicating degradation capture is independent of temperature scale. The 25°C band holds the most cycles (236) and widest SoH coverage, yielding the lowest R^2 (0.85) from greater target variance. The validation MCU’s SoH trajectory with predicted points is shown in Fig. 4.

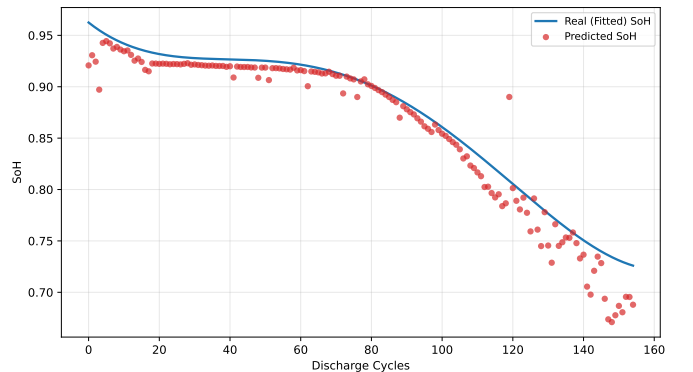


Fig. 4. Validation MCU (MCU3) SoH trajectory with predicted SoH points.

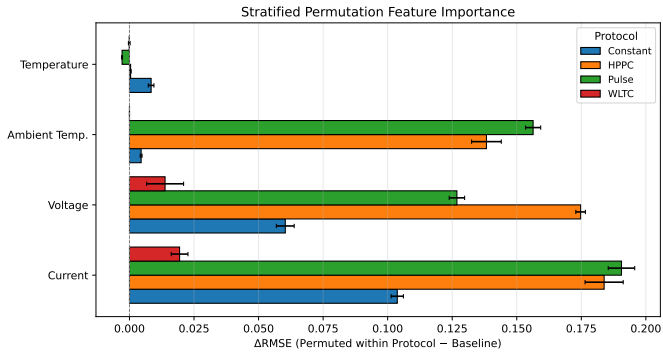


Fig. 5. PFI importance over the input features (V , I , T , T_{amb}).

TABLE IX
STRATIFIED PERMUTATION FEATURE IMPORTANCE FOR SoH ESTIMATOR

Feature	Constant	HPPC	Pulse	WLTC
Current	+0.1037	+0.1838	+0.1906	+0.0194
Voltage	+0.0603	+0.1747	+0.1268	+0.0138
Ambient Temp.	+0.0045	+0.1383	+0.1563	+0.0000
Temperature	+0.0084	+0.0004	-0.0029	-0.0001